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Galactic panel

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Eclipse Time Variations: third bodies or applegate mechanism?

Abstract

Several binary systems containing either a white dwarf or subdwarf and an M-type companion have exhibited time eclipse variations (ETVs). Previous studies suggest that these variations could be attributed to the presence of circumbinary planets. However, predictions based on ETVs have often failed in multiple systems. Additionally, all M-type stars in these binaries are magnetically active, necessitating a reevaluation of the Applegate mechanism (where magnetism induces stellar surface distortions, i.e. oblateness). One of the key predictions of the Applegate mechanism is the effective temperature change. Therefore, we propose to observe 2 of the 17 known systems visible during the semester to (1) monitor the ETVs and (2) measure the properties of the M-type stars (specifically their effective temperature).

Observing Blocks

Instrument/Telescope	Req. time	Min. time	1 st Option	2 nd Option
CCD Specu- loos/Speculoos	4 nights	3 nights	August Any	September Any

Cols

Name	Institution	e-mail	Observer?
Maja Vuckovic	UV	maja.vuckovic@uv.cl	False
monica zorotovic	UV	monica.zorotovic@uv.cl	False
Pia Amigo	USM	pia.amigo@usm.cl	False

Status of the project

- Past nights: 0
- Future nights: 12
- Long term: False

- Large program: False
- Thesis: False

List of Targets

ID	RA	DEC	Mag
HS 2231+2441	22:34:21.48	+24:56:57.39	G=14.17
NSVS 14256825	20:20:00.46	+04:37:56.52	G=13.22

Astrophysical context: Post-Common Envelope Binaries (PCEBs) are binary systems consisting of a primary component that is either a hot subdwarf (of spectral type B/O, hereafter sdBO) or a white dwarf (WD), paired with a secondary component that is a low-mass main sequence (MS) star or a brown dwarf (BD), with a small separation between the components. A widely accepted formation scenario for this configuration involves the evolution of the more massive star into a red giant, which then fills its Roche lobe. This results in a rapid and dynamically unstable mass transfer to the companion, leading to the formation of a common envelope (CE). The drag forces between the common envelope and the two stars cause them to spiral inward until sufficient energy is transferred to the CE, ultimately ejecting it from the system, leading to the compact PCEB configuration that we can observe today (Paczynski, 1976; Webbink, 1984; Zorotovic et al., 2010). In addition, PCEBs are progenitors of many interesting astrophysical objects, including cataclysmic variables (CVs), Type Ia supernovae, or even gravitational-wave sources (Anguiano et al., 2022). This makes PCEBs fundamental objects for studying stellar parameters and binary evolution. One particular *eclipsing* PCEB is NN Ser, which has attracted lots of attention given the presence of eclipse time variations (ETVs), and hence it has been largely studied (see for example Qian et al., 2009; Beuermann et al., 2010; Parsons et al., 2014; Özdönmez et al., 2023, and references therein). Furthermore, Hardy et al. (2016), based on ALMA observations, detected insights of dust surrounding the system. While NN Ser has been largely scrutinized, it is far from being the only system where ETVs have been detected (see Faillace et al., 2020; Han et al., 2018; Almeida et al., 2020, for examples).

Competing models: These ETVs detected in several PCEBs can be explained typically by several mechanisms. The most exciting one is the presence of a third (or more) sub-stellar object periodically moving the centre of mass of the binary system, although the formation mechanism of such objects is not yet fully understood. One scenario is that these objects formed with the binary system (first generation) and they survived and remain in stable orbits throughout the CE evolution. However, not many systems are expected to have survived (Mustill et al., 2013). In the second scenario, circumbinary objects formed from the remaining material after the envelope expulsion (Zorotovic & Schreiber, 2013) as a second generation planets. However, this requires an exceptionally efficient planet formation process, with more than 20% of the disc material converting into planets (Bear & Soker, 2014). Another mechanism associated with ETVs is the Applegate mechanism (Applegate, 1992), which suggests that cyclic changes in the magnetic activity of one of the stars leads to periodic distortions of the gravitational quadrupole moment (i.e. the oblateness) of the MS star and subsequent changes in the system’s gravitational dynamics, linked to variations in the orbital period.

The Applegate mechanism produces a modulation of the period with timescales of decades; conversely, the presence of a third body can produce a wide range of timescales depending on its orbital period. Thus, in order to identify the appropriate mechanism that causes the ETVs, it is crucial to obtain precise eclipse timings and well-characterized light curves of these systems.

For example, in the case of NN Ser, the detected ETVs suggests the present of two circumbinary objects with minimum masses of 6.9 and $2.2M_j$ (although Özdönmez et al. 2023 provide evidence ruling out the second planet and rising the minimum mass of the first planet to $9.5M_j$). Further analyses (cited above) have also ruled out other possible causes of the orbital period changes, notably the Applegate mechanism (Parsons et al., 2014). However, all these studies have not been able to make a compelling case for the the presence of circumbinary planet(s). Lately, Faillace et al. (2020) reported a departure from the historical O-C diagrams for several systems studied before, detecting a variation of a few tens milliseconds in the orbital period. Figure 1 shows the cases of NN Ser (left) and HS 0705+6700 (right). They concluded that additional observations over a longer time-line are needed in order to confirm this scenario or to uncover further period change complexities.

Roadmap for understanding the cause of ETVs: Persevering in ETVs monitoring The headcount of PCEBs which present ETVs are 17 systems (Parsons et al., 2010; Zorotovic & Schreiber, 2013; Wolf et al., 2021). To determine ETVs with high precision, it is imperative that the ingress and egress of the eclipse are well-sampled (see Figure 2). Among these systems the primary has been identified as a WD or a sdO while all the companions are M-type star with a large fraction of them showing signs of activity (e.g. flares; Parsons et al., 2010). In some of the systems, the stars are interacting via mass transfer due to Roche lobe overflow of the secondary (Zorotovic & Schreiber, 2013). In all these interacting systems in which ETVs have been detected, the WD is magnetic (Biermann et al., 1985; Berriman & Smith, 1988; Hakala et al., 1993). Given the strong signs of the presence of magnetism, and in addition to the persistent failures in the predictions made assuming the presence of planets around these binaries, it is natural to revisit the models. The Applegate mechanism can be tested in some ways. First, the brightness of the star should change in sync with the changes in its orbit. Second, other signs of magnetic activity, like starspots, X-ray brightness, or specific emissions in certain light wavelengths, should also show this same pattern. The changes in brightness are caused by the temperature changes, not because the star is getting bigger or smaller. When the star is the brightest, it should also be the hottest and bluest in color (Applegate, 1992). Hence, the light curves serve as invaluable datasets for deriving the stellar parameters, particularly the effective temperature of the secondary star. Given the close proximity of the secondary star to the significantly hotter WD/sdO, it experiences substantial irradiation on the side facing the primary. Consequently, accurate determination of the effective temperature requires meticulous modelling of the reflection effect. This phenomenon manifests as an additional modulation superimposed on the eclipses, with a more pronounced impact observed in redder filters. Shown in Figure 2, the light curves of NN Ser exemplify this phenomenon, wherein the r' filter exhibits a more pronounced reflection effect because the r' light is dominated by the secondary star (see figure 3). In essence, it is crucial to keep an eye over both ETVs and potential atmospheric changes in the secondary star.

Immediate objectives: The immediate goals of this proposal are twofold: a) to measure orbital periods from well-sampled light curves for two PCEB systems with known ETVs and b) to determine precise stellar parameters of the secondary component. To achieve our goals, we request four nights of observations with SPECULOOS to obtain multi-band lightcurves of two PCEB systems. The systems were selected from the 17 PCEB systems to meet the instrument capabilities and the observation window for the observation period. For the analysis, we will extract the differential light curves using the PROSE code (Garcia et al., 2022) and determine the orbital periods utilizing appropriate methods, such as Lomb-Scargle periodograms or PERIOD04. From the light curves we will be able to time up the mid-eclipses in order to extend the time baselines of the O-C diagrams. In addition, we will model the obtained lightcurve using PHOEBE (Prša et al., 2016), which will provide important parameters of both the stellar components (masses and effective temperatures) and the configuration of the binary (semi-major axis, inclination, eccentricity, argument of the periastron, etc.). Using PHOEBE we will also be able to detect signs of interaction between the stars, such as Roche distortion and irradiation from the hotter star to the secondary component. This complete characterization of our target binary systems is crucial to disentangle the possible models (Applegate mechanism and / or presence of circumbinary objects) that are able to explain the observed ETVs.

References

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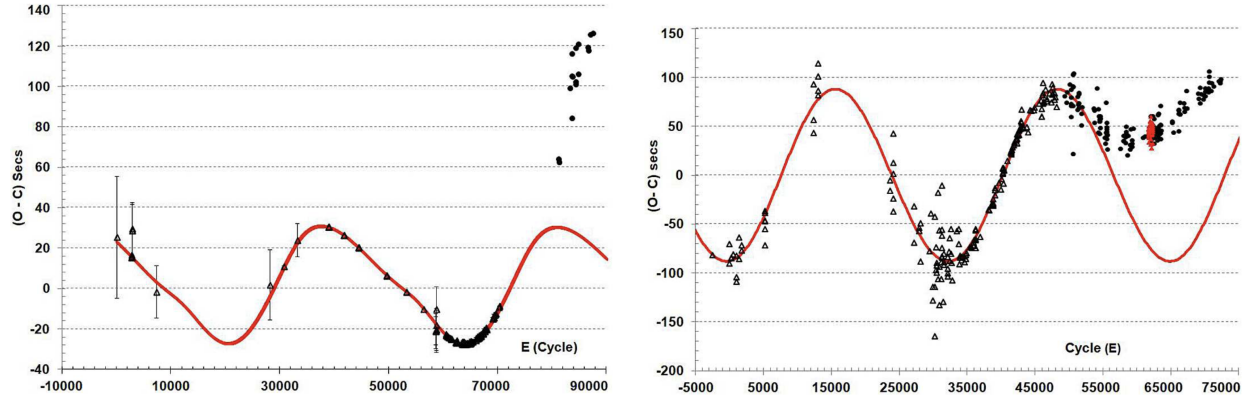


Figure 1: *Left:* NN Ser the first discovered PCEB displaying ETVs. For a while their eclipse time variations behave accordingly to the predictions, however the latest measurements show discrepancies, *Right:* HS0705+6700, a PCEBs with a hot subdwarf primary star which also shows deviations of the O-C predictions. Figures from [Faillace et al. \(2020\)](#).

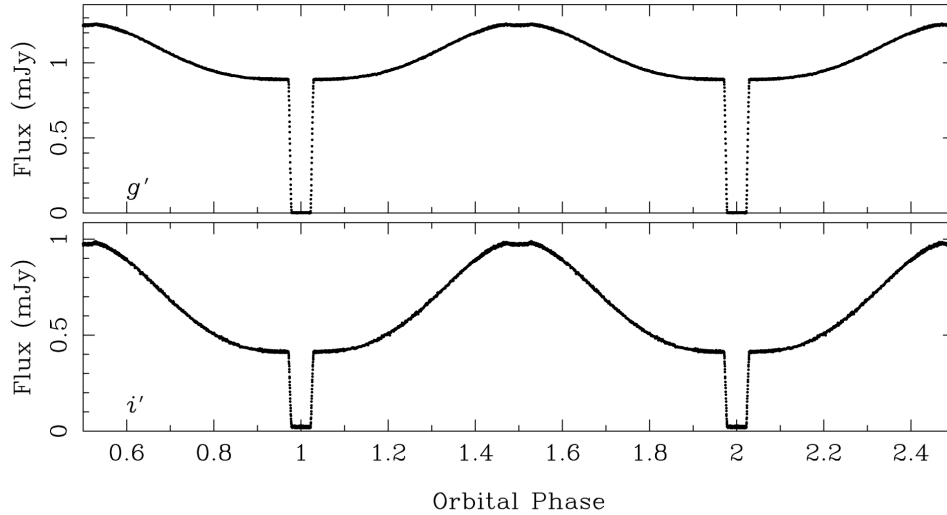


Figure 2: NN Ser lightcurve taken with ULTRACAM/La Silla using filters g' (top) and i' (bottom). The i' lightcurve shows a larger flux amplitude due to a strong reflection effect ([Parsons et al., 2010](#)).

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TECHNICAL DESCRIPTION

Our targets (table 1) have optical lightcurves observed with *TESS*. However, the single-band lightcurve data available from *TESS* is not sufficient for comprehensive modelling. SPECULOOS is the telescope equipped with the necessary aperture for high cadence observations and the required filters for the proposed science. We have used the SPOCK Chilean pipeline introducing representative template of a compact object plus M dwarf companion (See figure 3). Our targets are relatively bright allowing exposure times of less than 15 seconds (see table 1). The read-out plus overhead time is only $\simeq 10$ s. The combined short times will allow the temporal sampling needed. The time resolution on the lightcurves, will allow to identify the ingress and egress of the eclipses. Both targets should get at least two primary eclipses per night, which allows to phase fold the lightcurve, and thus to calculate the orbital period of the targets. In order to perform the PHOEBE modelling which include the modulation caused by the reflection light from the secondary, we need to have lightcurves at different filters (see Figure 2). Thus, we request the filters g' and r' that fully cover the Balmer lines (β and γ) from the white dwarf/subdwarf and the secondary. Since the readout plus overhead is very short (only 10 seconds), we will not request binning. We will make use of the standard calibrations. We request observations for *two* targets in *two* filters (g' and r') for *one* night each target. In total we request four nights of observation with SPECULOOS this 2025A semester.

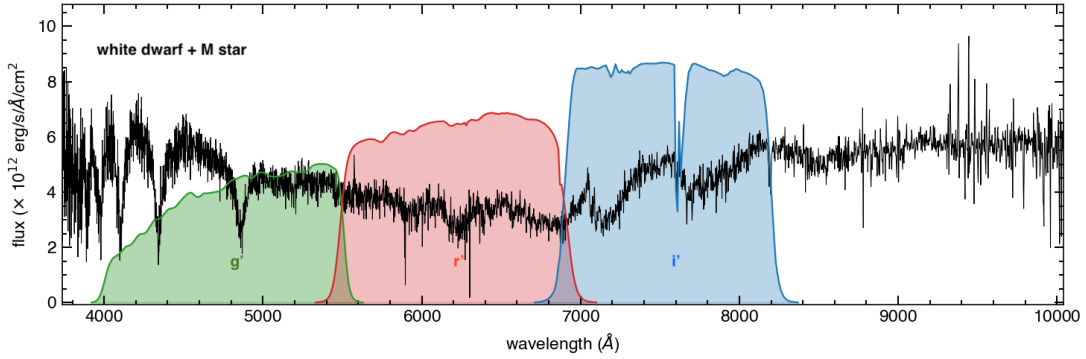


Figure 3: In the g' band, the signal encompasses $H\beta$ and is predominantly dominated by the flux emitted by the white dwarf. Conversely, in the r' band, the signal is primarily dictated by the flux from the secondary star. As a result, the slope of the lightcurve varies between the primary and secondary eclipses (see figure 2)

Table 1: Targets of compact binry displaying orbit changes. **sdO+M**: corresponds to a subdwarf as the compact primary component and a M companion.

Name	binary type	P_{orb} (days)	primary mass ($M_{\odot}$)	secondary mass ($M_{\odot}$)	RA	DEC	G (mag)	pref. date	T_{exp} (s)	<i>TESS</i>
HS2231+2441	sdO+M	0.110588	0.47	0.075	338.59	24.95	14.17	aug	30	✓
NSVS14256825	sdO+M	0.110374	0.419 ± 0.070	0.109 ± 0.023	305.00	4.63	13.22	aug	15	✓

CURRENT STATUS OF THE PROJECT

In general, the modelling of the O-C diagram for these system assume a quadratic term (e.g. dashed red line in the right panel in Figure 4), i.e

$$T_{\min, \text{BJD}} = T_0 + P_{\text{orb}} E + f E^2 + \tau$$

where E is the number of binary periods measured from the reference epoch T_0 , f a constant and τ is the cyclical light-time travel effect of a putative circumbinary object, given by:

$$\tau = \frac{a_{12} \sin i}{c} \left[(1 - e^2) \frac{\sin(v + w)}{1 + e \cos v} + e \sin w \right]$$

and $(a_{12} \sin i) / c$ is the light-time travel amplitude; e is the circumbinary component orbital eccentricity; w is the angle of periastron, and v is the true anomaly, determined from the time at periastron and the circumbinary period. However, this quadratic model fails to make accurate predictions over a longer time baseline. Several systems have already shown deviations from the predicted values (e.g. [Faillace et al., 2020](#)).

Out of the sample of 17 systems, 15 systems have been observed with *TESS*. From these, we extracted a light curve from a single pixel (see left panel in Figure 4) and verified that the period corresponds to the PCEB using *TESS_LOCALIZE*. We obtain the mid-eclipses from Gaussian fits to the different *TESS* sectors (see right panel in Figure 4), which provides > 50 time eclipses in each sector. However, since the data is available only in a broad-band filter, modeling the light curve using solely *TESS* data is insufficient to accurately determine the secondary stellar parameters. *Therefore, the multi-band lightcurve will allow the O-C monitoring, as well as the characterisation of the secondary.*

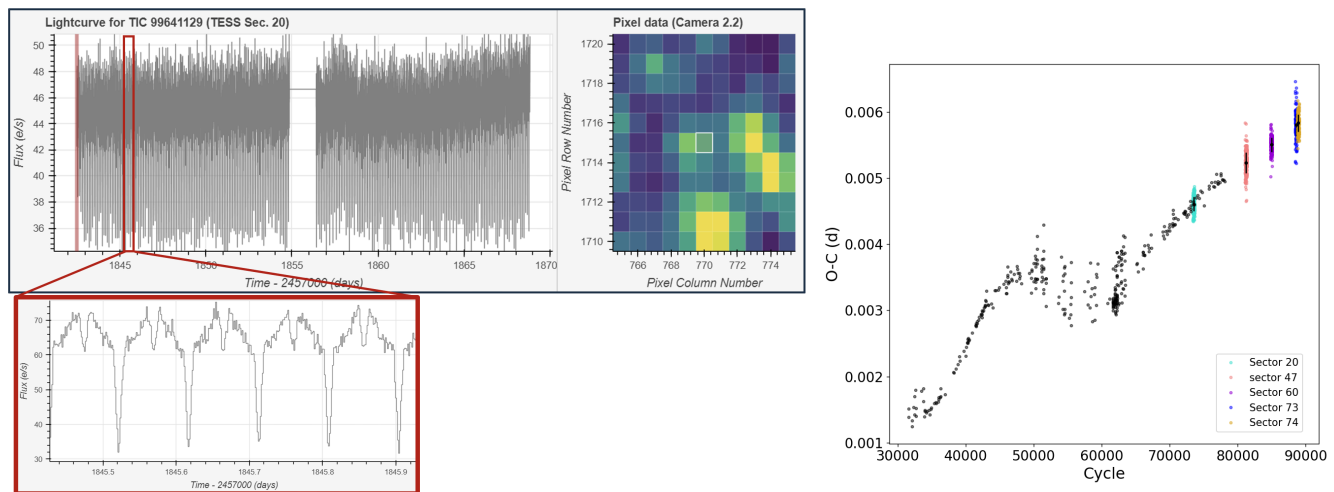


Figure 4: *Left*: *TESS* lightcurve extracted from one pixel at the target position, which clearly shows several primary and secondary eclipses (zoomed-panel). *Right*: O-C diagram for the PCEB HS25+6700. The mid-eclipses from the literature in black dots ([Faillace et al., 2020](#); [Bogensberger et al., 2017](#); [Qian et al., 2013](#); [Pulley et al., 2018, 2015](#); [Beuermann et al., 2012](#); [Pulley et al., 2022](#)) and from *TESS* sectors 20 (orange), 47 (blue), 47 (green), 73 (magenta) and 74 (cyan), which the eclipse time measurements are averaged.